

Software Requirements Specification — KNMP VisitHub

Document Title: Software Requirements Specification (SRS) **System:** KNMP VisitHub — A Web-Based Booking and Visit Planning System for the Kwame Nkrumah Memorial Park **Author:** Manus AI **Date:** August 2026 **Revision:** 1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of KNMP VisitHub, a web-based booking and visit-planning system for the Kwame Nkrumah Memorial Park in Accra, Ghana. It is intended for the project's stakeholders, developers, testers, and maintenance engineers, and serves as the acceptance baseline against which the delivered system can be verified.

1.2 Scope

KNMP VisitHub provides visitors with a public interface for exploring attractions, viewing fees, making dated bookings with automatic cost calculation, receiving unique booking references and PDF tickets, and maintaining personal visit itineraries. It provides administrators with a protected dashboard for managing attractions, visitor fee categories, bookings and their statuses, user accounts, park-wide operational settings, and analytics. Gate staff use a dedicated check-in page to scan ticket QR codes and record arrivals.

1.3 Definitions and Acronyms

Term	Definition
KNMP	Kwame Nkrumah Memorial Park, Accra, Ghana
Pesewas	Ghanaian subunit of the Cedi (100 pesewas = 1 Cedi); all prices are stored as integer pesewas
Manus OAuth	The authentication service used for user sign-in
tRPC	End-to-end type-safe remote procedure call framework
TiDB	MySQL-compatible distributed database (serverless)
Check-in	The act of recording a booked party's arrival at the park gate

1.4 References

The requirements in this document derive from the user's project brief, which specified the system aim, eleven project objectives, and two actor profiles (Visitor and Administrator).

2. Overall Description

2.1 User Characteristics

Visitor. The primary public user. Technically literate, uses the system to plan and book park visits. Requires no training beyond the in-application interface.

Administrator. Park staff responsible for content, bookings, users, and reporting. Requires familiarity with web dashboards but no specialized training.

Gate staff. Operational staff who validate tickets at park entrances using the check-in page; their workflow must be fast and tolerant of offline camera conditions.

2.2 Operating Environment

The system runs as a browser-accessible web application (desktop and mobile browsers). No client installation is required. The server operates on autoscaling serverless hosting with a TiDB Serverless database and the Resend transactional email service.

2.3 Design and Implementation Constraints

Authentication must use Manus OAuth; local password accounts are not permitted for sign-in (administrators can pre-register accounts by email for matching at first OAuth sign-in). Monetary amounts are integer pesewas. All business timestamps are stored as UTC values. The system must degrade gracefully when the email provider is unavailable: bookings and status updates must never fail because email delivery fails.

2.4 Assumptions and Dependencies

It is assumed that visitors possess valid email addresses for ticket delivery and that park staff have network-connected devices capable of camera access for QR scanning. The system depends on the availability of Manus OAuth, the TiDB database region, and the Resend API.

3. Specific Requirements

3.1 Functional Requirements

FR-1 Attractions.

- FR-1.1 The system shall display all KNMP attractions with name, description, images, location, category, and opening hours.
- FR-1.2 The system shall allow visitors to search attractions by keyword.
- FR-1.3 The system shall allow visitors to filter attractions by category and location, showing live counts per facet.

FR-2 Fees.

- FR-2.1 The system shall display applicable visitor categories and per-category fees.
- FR-2.2 The booking form shall compute the total cost live as quantities change.

FR-3 Bookings.

- FR-3.1 The system shall generate a unique reference of the form `KNMP-XXXXXX` for every booking.
- FR-3.2 Visitors shall select a visit date or a multi-day date range.
- FR-3.3 Visitors shall specify visitor categories and quantities.
- FR-3.4 The system shall validate that a party does not exceed daily capacity constraints when warnings are active.

- FR-3.5 Visitors shall view, cancel (owner-only), and download a PDF ticket for each booking.
- FR-3.6 The system shall email the PDF ticket to the visitor's registered address on successful booking.
- FR-3.7 Guided-tour time slots shall be selectable per attraction per day.

FR-4 Itineraries.

- FR-4.1 Signed-in visitors shall create and manage a personal visit itinerary (add, edit, reorder, remove items with planned times and notes).
- FR-4.2 Visitors shall export their itinerary as a PDF.
- FR-4.3 Visitors shall generate a shareable public link to their itinerary.

FR-5 Authentication and Accounts.

- FR-5.1 All protected features shall require Manus OAuth sign-in.
- FR-5.2 The system shall reject inactive (blocked) users from all protected endpoints.
- FR-5.3 Administrators shall list registered users, change roles, and activate or deactivate accounts; an administrator shall not be able to demote themselves.
- FR-5.4 Administrators shall create new user accounts by name, email, and role, matched to the user's OAuth email at first sign-in.

FR-6 Administration.

- FR-6.1 Administrators shall create, edit, and delete attractions (including images and opening hours).
- FR-6.2 Administrators shall create, edit, and delete visitor categories and prices.
- FR-6.3 Administrators shall view all bookings, update booking status (pending/confirmed/cancelled), and view booking details including line items and tour slots.
- FR-6.4 Changing a status to confirmed or cancelled shall email the visitor (falling back to the account email), never blocking the status update on email failure.
- FR-6.5 Administrators shall configure daily visitor capacity and the near-capacity warning threshold via the Settings page.
- FR-6.6 Gate staff shall scan ticket QR codes (camera) or enter references manually to check parties in, with an undo capability.
- FR-6.7 The bookings list shall visually indicate checked-in parties and provide an All / Checked in / Not checked in filter.

FR-7 Analytics and Reporting.

- FR-7.1 Administrators shall view KPIs (bookings, visitors, revenue), a visitor-category breakdown with CSV export, and monthly revenue trends.
- FR-7.2 Analytics shall be filterable by an arbitrary visit-date range.
- FR-7.3 The system shall display a 14-day daily visitor-capacity forecast derived from booked quantities (multi-day stays spread evenly), with a capacity reference line.

FR-8 Audit.

- FR-8.1 The system shall record an immutable audit event for role changes, account activations/deactivations, attraction and category CRUD operations, booking-status changes, gate check-ins and undos, and user creation.
- FR-8.2 Only administrators shall access the audit log.

3.2 Non-Functional Requirements

NFR-1 Performance. The dashboard and explorer pages shall render within 2 seconds on a broadband connection; analytics queries shall complete within 10 seconds. The check-in lookup shall complete within 3 seconds to avoid gate queues.

NFR-2 Security. Passwords are never stored; authentication is fully delegated to Manus OAuth. Administrative actions require the admin role. Visitor data is owner-scoped. Secrets are managed outside the codebase as deployment environment variables.

NFR-3 Reliability. Email outages must not prevent bookings or status updates. Database failures surface typed errors rather than hanging requests.

NFR-4 Usability. The interface shall be responsive across desktop and mobile viewports, use a consistent design system, and provide loading, empty, and error states for all data views.

NFR-5 Maintainability. The codebase is fully typed (TypeScript), with server and client sharing one type contract through tRPC, and is covered by an automated test suite.

NFR-6 Portability. The application is browser-based and requires no client installation; deployment is container-compatible via the hosting platform.

4. Use Case Model

ID	Use Case	Actor	Primary Flow
UC-1	Browse and filter attractions	Visitor	Open explorer → search/apply facets → view details
UC-2	Make a booking	Visitor	Choose dates → pick categories and quantities → review live cost and capacity banner → confirm → receive reference and PDF email
UC-3	Manage my bookings	Visitor	View list → download ticket → cancel if pending
UC-4	Build itinerary	Visitor	Create itinerary → add attractions with times/notes → reorder → export PDF or share link
UC-5	Manage attractions and categories	Administrator	CRUD through dashboard forms; audits recorded
UC-6	Process bookings	Administrator	Update status (emails visitor) → view details → gate check-in via QR/manual
UC-7	Gate check-in	Gate staff	Scan QR or enter reference → party marked checked in → undo if mistaken
UC-8	Review analytics	Administrator	Apply date range → review KPIs, breakdown (CSV export), trends, and forecast
UC-9	Manage users	Administrator	List users → change role → activate/deactivate → create new account
UC-10	Configure settings	Administrator	Set daily capacity and warning threshold; applied across forecast and public booking form

5. External Interface Requirements

The user interface follows the React 19 + Tailwind 4 component system with shadcn/ui primitives. The system interface comprises the tRPC gateway (`/api/trpc`), the OAuth callback endpoint (`/api/oauth/callback`), and public share routes (`/share/itinerary/:code`). The hardware interface concerns camera access on staff devices for QR scanning, requested at runtime and degradable to manual entry.

6. Acceptance Criteria

The system is accepted when: all eleven project objectives are demonstrated; every functional requirement in Section 3.1 is verifiable through the running application; the automated suite (77 tests across six files) passes; and the deployment serves the full visitor and administrator workflows without intervention.